SKE

SMARTKEY POWER

Precision Air Conditioning / Micro-Module

Your reliable energy partner worldwide







Base-station AC | Room-level AC | In-row AC | Rack-mount AC

Company profile

COMPANY PROFILE

Shenzhen Smartkey Power Co., Ltd. (registered capital RMB 50 million) is a national high-tech enterprise and a Shenzhen “specialized and innovative” enterprise that develops, manufactures and sells UPS systems. Its main products include off-line UPS, on-line UPS (1–800 kVA) and DC UPS. Building on these, it has developed rack-mount UPS, rack-mount inverters, modular UPS, UPS for projectors, medical and surveillance applications — more than 100 UPS solutions tailored to markets around the world.

SKE operates 30,000 m² of production and R&D facilities in Shenzhen and Dongguan with an advanced production management system, and is committed to in-house R&D and brand building. Its products hold CE, TLC, energy-saving and other domestic and international certifications and are exported to more than 120 countries and regions. Service centres in 7 regions of China form a network covering the country’s main cities and counties, ensuring fast, efficient service. With strong R&D and a quality service platform, SKE has earned a good reputation in steel and machinery, metallurgy, petrochemicals, ports, oil and gas, power, banking, broadcasting, healthcare and many other sectors.

SKE places great emphasis on scientific human-resource management and has built a training-oriented talent development system — one of the few companies in China’s UPS industry to make training part of daily management. Guided by the values “customer first, pragmatism, efficiency, innovation and mutual benefit” and the principle “everything centred on the customer”, SKE is dedicated to providing customers worldwide with the best power solutions and service.







Steady as a mountain, with you every moment

Product range



Base-station air conditioner (single circuit, 8–20 kW)

For telecom base stations · fixed-speed rotary/scroll · 8 models

8–20 kW



Room-level precision AC (single circuit, 25–50 kW)

Data rooms · up-flow/down-flow · 10 models

25–50 kW



Room-level precision AC (dual circuit, 50–100 kW)

Medium/large data centres · dual circuit · 12 models

50–100 kW



In-row precision AC (single circuit, 12.5–50 kW)

High heat density · full inverter · 14 models

12.5–50 kW



Rack-mount precision AC (single circuit, 3.5–12.5 kW)

Integrated cabinets / micro-modules · 6 models

3.5–12.5 kW

Base-station air conditioner (single circuit, 8–20 kW)

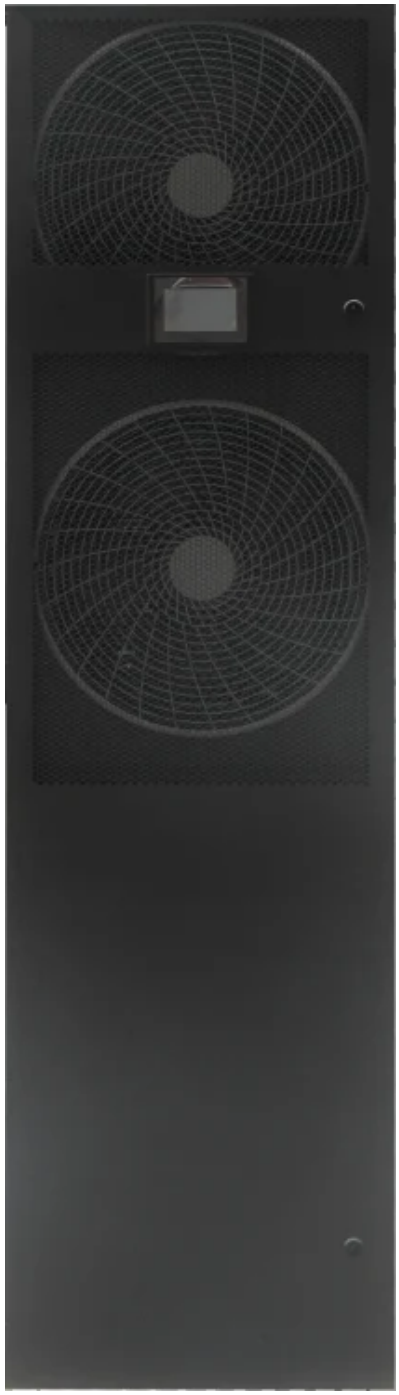
Overview

The FAJ series is a dedicated air conditioner developed for mobile communication base stations. It gives high-tech electronic equipment maximum reliability, top operating efficiency, stable temperature control and good energy savings. The units run stably and efficiently with a low failure rate. Cooling capacities from 8 to 20 kW meet temperature and humidity control needs in many different sites.

Applications

· Data rooms
· Small and medium data centres
· Substations and switch rooms
· Storage and testing rooms
· 5G base stations
· Equipment rooms needing precise temperature & humidity control

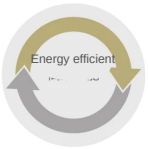






Base-station AC

Features



Energy efficient

· High airflow, small enthalpy difference and a high sensible heat ratio suit equipment rooms, providing a continuous, stable temperature and humidity environment.
· High-efficiency compressor and axial fan with precise electronic expansion valve control maximise efficiency.
· Stepless outdoor-fan speed control follows condensing pressure to cut overall energy use.
· Group control manages several units centrally for whole-room energy savings.
· R410A refrigerant as standard — environmentally friendly and efficient.



Stable and reliable

· Components from well-known brands; stable, reliable component and system design.
· Comprehensive multi-level alarms and protection: power, water leakage, temperature/humidity limits and component protection.
· Stable operation over a very wide input voltage range, with phase-sequence detection and other power protection.
· Automatic restart after power returns.



Intelligent control

· 4.3-inch colour touch screen shows system status and component operation in real time.
· Multi-level password protection; alarms are graded automatically by severity.
· Group control with rotation and standby: duty and standby units switch over automatically at set times.
· Standard RS485 monitoring interface; standard Modbus protocol provided free of charge.
· Remote setup and monitoring of unit status and alarms.



Flexible

· Threaded quick couplings as standard — fast installation on site, no brazing.
· Air supply arrangement matched to the site.
· Optional auxiliary heating or humidification as needed.



Base-station AC

High-efficiency components



High-efficiency DC axial fan

Speed is set by adjusting the voltage, with fast response, low noise, high efficiency and energy savings.



High-efficiency rotary compressor

Premium-brand rotary compressor with no high/low-pressure valves and liquid-slugging protection; low starting current and smooth, continuous, quiet operation with a long service life.



Electronic expansion valve

Stepless opening from 10% to 100%, working with the compressor for precise cooling: fast response, wide range, quiet and efficient. The valve is reversible and controls flow automatically in both cooling and heating.



High-efficiency humidifier

Intelligent humidifier control with a cleanable cylinder: adjustable output, automatic draining and flushing, suits different water qualities, saves energy and water, hygienic.



High-efficiency heater

High heat density in a compact design: fast, stable heating, low surface temperature and long life. Built-in power staging selects the heating level from real-time conditions to hold the environment precisely.



Technical specifications

Base-station AC · cooling only – Indoor unit

Indoor unit model	SK-FAJ008F	SK-FAJ013F	SK-FAJ017F	SK-FAJ020F
System type	Single circuit, fixed speed			
Air supply	Top-front supply			
Compressor	Fixed-speed rotary		Fixed-speed scroll	
Refrigerant	R410A			
Fan	DC axial fan			
Expansion valve	Electronic expansion valve			
Heater type	—			
Humidifier type	—			
Performance				
Total cooling capacity (kW)	8.2	13.5	17.5	20.5
Sensible cooling capacity (kW)	7.54	12.4	16.1	18.8
Sensible heat ratio	0.92			
Airflow (m³/h)	2400	3900	5500	6000
Heating capacity (kW)	—			
Humidification (kg/h)	—			
Condensate drain (mm)	20			
Pipe – gas line (mm)	12.7	16		
Pipe – liquid line (mm)	9.52		12.7	
Power supply (V/Hz)	380 V / 50 Hz			
Temperature control accuracy	±1 °C			
Humidity control accuracy	±5%			
Filter class	G2			
Physical				
Weight (kg)	95	110	162	166
W × D × H (mm)	510 × 425 × 1860		580 × 460 × 1950	

Base-station AC · cooling only – Outdoor unit

Indoor unit model	SK-FAJ008F	SK-FAJ013F	SK-FAJ017F	SK-FAJ020F
Outdoor unit model	SK-CY0121	SK-CY0191	SK-CY0241	SK-CY0291
Quantity	1			
Electrical & piping (example: within 30 m)				
Power supply (V/Hz)	220 V / 50 Hz			
Max. horizontal distance	30 m (with extension kit)		45 m (with extension kit)	
Max. height difference (outdoor unit above)	10 m			
Max. height difference (outdoor unit below)	5 m			
Physical				
Dimensions (mm)	960 × 370 × 810	960 × 370 × 1260	1060 × 380 × 1429	
Weight (kg)	35	53	90	100



Technical specifications

Base-station AC · constant temperature & humidity – Indoor unit

Indoor unit model	SK-FAJ008FRH	SK-FAJ013FRH	SK-FAJ017FRH	SK-FAJ020FRH
System type	Single circuit, fixed speed			
Air supply	Top-front supply			
Compressor	Fixed-speed rotary		Fixed-speed scroll	
Refrigerant	R410A			
Fan	DC axial fan			
Expansion valve	Electronic expansion valve			
Heater type	PTC			
Humidifier type	Electrode			
Performance				
Total cooling capacity (kW)	8.2	13.5	17.5	20.5
Sensible cooling capacity (kW)	7.54	12.4	16.1	18.8
Sensible heat ratio	0.92			
Airflow (m³/h)	2400	3900	5500	6000
Heating capacity (kW)	3			
Humidification (kg/h)	3			
Condensate drain (mm)	20			
Pipe – gas line (mm)	12.7	16		
Pipe – liquid line (mm)	9.52		12.7	
Power supply (V/Hz)	380 V / 50 Hz			
Temperature control accuracy	±1 °C			
Humidity control accuracy	±5%			
Filter class	G2			
Physical				
Weight (kg)	95	110	162	166
W × D × H (mm)	510 × 425 × 1860		580 × 460 × 1950	

Base-station AC · constant temperature & humidity – Outdoor unit

Indoor unit model	SK-FAJ008FRH	SK-FAJ013FRH	SK-FAJ017FRH	SK-FAJ020FRH
Outdoor unit model	SK-CY0121	SK-CY0191	SK-CY0241	SK-CY0291
Quantity	1			
Electrical & piping (example: within 30 m)				
Power supply (V/Hz)	220 V / 50 Hz			
Max. horizontal distance	30 m (with extension kit)		45 m (with extension kit)	
Max. height difference (outdoor unit above)	10 m			
Max. height difference (outdoor unit below)	5 m			
Physical				
Dimensions (mm)	960 × 370 × 810	960 × 370 × 1260	1060 × 380 × 1429	
Weight (kg)	35	53	90	100

Room-level precision AC (single circuit, 25–50 kW)

Overview

FAJ series precision air conditioners provide constant temperature and humidity control. High airflow, a high sensible heat ratio and high energy efficiency meet the needs of professional machine rooms: the units are designed for 24/7, 365-day operation, with a precisely matched high-efficiency refrigeration system and a large evaporator for optimum energy-saving operation around the clock. Cooling capacities from 25 to 50 kW cover temperature and humidity control needs in a wide range of sites.

Applications

· Data rooms
· Small and medium data centres
· Substations and switch rooms
· Storage and testing rooms
· Equipment rooms needing precise temperature & humidity control





Features

Modular design

· Modular design: the unit can be quickly taken apart and reassembled on site — where lift doors or corridors are narrow, it splits into several modules that pass through easily.
· Full front-access maintenance, with side and rear access too; opening the front door of each module reaches every component.



High-efficiency design

· Up-flow and down-flow units use high-efficiency “V/A” coil heat exchangers: larger heat-exchange and filter areas improve heat transfer while lowering air resistance and fan power.



Intelligent control

· Colour touch screen shows system status and component operation in real time.
· Multi-level password protection; alarms are graded automatically by severity.
· Group control with rotation and standby: duty and standby units switch over automatically at set times.
· Standard RS485 monitoring interface; standard Modbus protocol provided free of charge.
· Remote setup and monitoring of unit status and alarms.



Excellent electrical performance

· Reliable operation at outdoor temperatures of −15 °C to 45 °C; down to −35 °C with the optional low-temperature kit.
· Breakers and contactors from internationally known brands; wide-range supply 380 V ±15%, 3N~, 50 Hz ±2 Hz.



High-efficiency components



High-efficiency EC fan

Compared with belt-driven forward-curved fans, backward-curved centrifugal fans mean lower power, less maintenance and better airflow, improving cooling efficiency. DC variable-speed EC fans with up to 90% efficiency offer a wide airflow range and lower energy use.



High-efficiency scroll compressor

Brand-name scroll compressor with no high/low-pressure valves and liquid-slugging protection; low starting current, smooth and quiet with a long service life. The cabinet is made by CNC punching and dedicated tooling with one-piece panels for high strength and heat resistance.



High-efficiency heater

High heat density in a compact design: fast, stable heating, low surface temperature and long life. Built-in power staging selects the heating level from real-time conditions to hold the environment precisely.



High-efficiency humidifier

Intelligent humidifier control with a cleanable cylinder: adjustable output, automatic draining and flushing, suits different water qualities, saves energy and water, hygienic.



Technical specifications

Room-level AC · cooling only – Indoor unit

Indoor unit model	SK-FAJ025U(D)	SK-FAJ030U(D)	SK-FAJ035U(D)	SK-FAJ040U(D)	SK-FAJ050U(D)
System type	Single circuit, fixed speed				
Air supply	Up-flow (U) / down-flow (D)				
Compressor	Fixed-speed scroll				
Refrigerant	R410A				
Fan	EC centrifugal fan				
Expansion valve	Electronic expansion valve				
Heater type	—				
Humidifier type	—				
Performance					
Total cooling capacity (kW)	25.4	30.5	35.4	40.2	50.5
Sensible cooling capacity (kW)	23.3	28.06	32.56	36.98	46.46
Sensible heat ratio	0.92				
Airflow (m³/h)	7000	8500	10000	11500	14200
Heating capacity (kW)	—				
Humidification (kg/h)	—				
Condensate drain (mm)	20				
Pipe – gas line (mm)	19			22	
Pipe – liquid line (mm)	16				
Power supply (V/Hz)	380 V / 50 Hz				
Temperature control accuracy	±1 °C				
Humidity control accuracy	±5%				
Filter class	G4				
Physical					
Weight (kg)	210	240	290	340	360
W × D × H (mm, excl. plenum)	930 × 970 × 1950				
Up-flow plenum height (mm)	450				

Room-level AC · cooling only – Outdoor unit

Indoor unit model	SK-FAJ025U(D)	SK-FAJ030U(D)	SK-FAJ035U(D)	SK-FAJ040U(D)	SK-FAJ050U(D)
Outdoor unit model	SK-CY0451	SK-CY0522	SK-CY0582	SK-CY0662	SK-CY0893
Quantity	1				
Electrical & piping (example: within 30 m)					
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz	
Max. horizontal distance	60 m (with extension kit)				
Max. height difference (outdoor unit above)	20 m				
Max. height difference (outdoor unit below)	5 m				
Physical					
Dimensions (mm)	1530 × 400 × 750	1530 × 400 × 996	1785 × 400 × 955	1850 × 400 × 996	2200 × 400 × 996
Weight (kg)	115	128	140	160	185



Technical specifications

Room-level AC · constant temperature & humidity – Indoor unit

Indoor unit model	SK-FAJ025U(D)RH	SK-FAJ030U(D)RH	SK-FAJ035U(D)RH	SK-FAJ040U(D)RH	SK-FAJ050U(D)RH
System type	Single circuit, fixed speed				
Air supply	Up-flow (U) / down-flow (D)				
Compressor	Fixed-speed scroll				
Refrigerant	R410A				
Fan	EC centrifugal fan				
Expansion valve	Electronic expansion valve				
Heater type	PTC				
Humidifier type	Electrode				
Performance					
Total cooling capacity (kW)	25.4	30.5	35.4	40.2	50.5
Sensible cooling capacity (kW)	23.3	28.06	32.56	36.98	46.46
Sensible heat ratio	0.92				
Airflow (m³/h)	7000	8500	10000	11500	14200
Heating capacity (kW)	6				
Humidification (kg/h)	6			8	
Condensate drain (mm)	20				
Pipe – gas line (mm)	19			22	
Pipe – liquid line (mm)	16				
Power supply (V/Hz)	380 V / 50 Hz				
Temperature control accuracy	±1 °C				
Humidity control accuracy	±5%				
Filter class	G4				
Physical					
Weight (kg)	230	260	310	360	410
W × D × H (mm, excl. plenum)	930 × 970 × 1950				
Up-flow plenum height (mm)	450				

Room-level AC · constant temperature & humidity – Outdoor unit

Indoor unit model	SK-FAJ025U(D)RH	SK-FAJ030U(D)RH	SK-FAJ035U(D)RH	SK-FAJ040U(D)RH	SK-FAJ050U(D)RH
Outdoor unit model	SK-CY0451	SK-CY0522	SK-CY0582	SK-CY0662	SK-CY0893
Quantity	1				
Electrical & piping (example: within 30 m)					
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz	
Max. horizontal distance	60 m (with extension kit)				
Max. height difference (outdoor unit above)	20 m				
Max. height difference (outdoor unit below)	5 m				
Physical					
Dimensions (mm)	1530 × 400 × 750	1530 × 400 × 996	1785 × 400 × 955	1850 × 400 × 996	2200 × 400 × 996
Weight (kg)	115	128	140	160	185

Room-level precision AC (dual circuit, 50–100 kW)

Overview

FAJ series precision air conditioners provide constant temperature and humidity control. High airflow, a high sensible heat ratio and high energy efficiency meet the needs of professional machine rooms: the units are designed for 24/7, 365-day operation, with a precisely matched high-efficiency refrigeration system and a large evaporator for optimum energy-saving operation around the clock. Cooling capacities from 50 to 100 kW cover temperature and humidity control needs in a wide range of sites.

Applications

· Data rooms
· Medium and large data centres
· Substations and switch rooms
· Storage and testing rooms
· Equipment rooms needing precise temperature & humidity control





Features

Modular design

· Modular design: the unit can be quickly taken apart and reassembled on site — where lift doors or corridors are narrow, it splits into several modules that pass through easily.
· Full front-access maintenance, with side and rear access too; opening the front door of each module reaches every component.



High-efficiency design

· Up-flow and down-flow units use high-efficiency “V/A” coil heat exchangers: larger heat-exchange and filter areas improve heat transfer while lowering air resistance and fan power.



Intelligent control

· Colour touch screen shows system status and component operation in real time.
· Multi-level password protection; alarms are graded automatically by severity.
· Group control with rotation and standby: duty and standby units switch over automatically at set times.
· Standard RS485 monitoring interface; standard Modbus protocol provided free of charge.
· Remote setup and monitoring of unit status and alarms.



Excellent electrical performance

· Reliable operation at outdoor temperatures of −15 °C to 45 °C; down to −35 °C with the optional low-temperature kit.
· Breakers and contactors from internationally known brands; wide-range supply 380 V ±15%, 3N~, 50 Hz ±2 Hz.

High-efficiency components



High-efficiency EC fan

Compared with belt-driven forward-curved fans, backward-curved centrifugal fans mean lower power, less maintenance and better airflow, improving cooling efficiency. DC variable-speed EC fans with up to 90% efficiency offer a wide airflow range and lower energy use.



High-efficiency scroll compressor

Brand-name scroll compressor with no high/low-pressure valves and liquid-slugging protection; low starting current, smooth and quiet with a long service life. The cabinet is made by CNC punching and dedicated tooling with one-piece panels for high strength and heat resistance.



Electronic expansion valve

A microprocessor-controlled valve meters refrigerant precisely, improving start-up, giving fast, stable control and better cooling. Its wide flow range lets the system make full use of low outdoor temperatures, saving up to 30% energy.



High-efficiency humidifier

Intelligent humidifier control with a cleanable cylinder: adjustable output, automatic draining and flushing, suits different water qualities, saves energy and water, hygienic.



High-efficiency heater

High heat density in a compact design: fast, stable heating, low surface temperature and long life. Built-in power staging selects the heating level from real-time conditions to hold the environment precisely.



Technical specifications

Room-level AC · cooling only – Indoor unit

Indoor unit model	SK-FAJ050U(D)-DS	SK-FAJ060U(D)	SK-FAJ070U(D)	SK-FAJ080U(D)	SK-FAJ090U(D)	SK-FAJ100U(D)
System type	Dual circuit, fixed speed					
Air supply	Up-flow (U) / down-flow (D)					
Compressor	Fixed-speed scroll					
Refrigerant	R410A					
Fan	EC centrifugal fan					
Expansion valve	Electronic expansion valve					
Heater type	—					
Humidifier type	—					
Performance						
Total cooling capacity (kW)	50.5	60.3	70.6	80.5	90.5	100.3
Sensible cooling capacity (kW)	46.46	55.48	64.95	74.06	83.26	92.28
Sensible heat ratio	0.92					
Airflow (m³/h)	15600	18400	21000	23000	25000	27000
Heating capacity (kW)	—					
Humidification (kg/h)	—					
Condensate drain (mm)	20					
Pipe – gas line (mm)	22					
Pipe – liquid line (mm)	16					
Power supply (V/Hz)	380 V / 50 Hz					
Temperature control accuracy	±1 °C					
Humidity control accuracy	±5%					
Filter class	G4					
Physical						
Weight (kg)	430	490	580	670	720	779
W × D × H (mm, excl. plenum)	1830 × 970 × 1950					
Up-flow plenum height (mm)	450					

Room-level AC · cooling only – Outdoor unit

Indoor unit model	SK-FAJ050U(D)-DS	SK-FAJ060U(D)	SK-FAJ070U(D)	SK-FAJ080U(D)	SK-FAJ090U(D)	SK-FAJ100U(D)
Outdoor unit model	SK-CY0522		SK-CY0582		SK-CY0752	SK-CY0742
Quantity	2					
Electrical & piping (example: within 30 m)						
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz		
Max. horizontal distance	60 m (with extension kit)					
Max. height difference (outdoor unit above)	20 m					
Max. height difference (outdoor unit below)	5 m					
Physical						
Dimensions (mm)	1530 × 400 × 996		1785 × 400 × 955		2035 × 400 × 996	
Weight (kg)	128		140	158	160	185



Technical specifications

Room-level AC · constant temperature & humidity – Indoor unit

Indoor unit model	SK-FAJ050U(D)RH-DS	SK-FAJ060U(D)RH	SK-FAJ070U(D)RH	SK-FAJ080U(D)RH	SK-FAJ090U(D)RH	SK-FAJ100U(D)RH
System type	Dual circuit, fixed speed					
Air supply	Up-flow (U) / down-flow (D)					
Compressor	Fixed-speed scroll					
Refrigerant	R410A					
Fan	EC centrifugal fan					
Expansion valve	Electronic expansion valve					
Heater type	PTC					
Humidifier type	Electrode					
Performance						
Total cooling capacity (kW)	50.5	60.3	70.6	80.5	90.5	100.3
Sensible cooling capacity (kW)	46.46	55.48	64.95	74.06	83.26	92.28
Sensible heat ratio	0.92					
Airflow (m^3/h)	15600	18400	21000	23000	25000	27000
Heating capacity (kW)	6					
Humidification (kg/h)	8					
Condensate drain (mm)	20					
Pipe – gas line (mm)	22					
Pipe – liquid line (mm)	16					
Power supply (V/Hz)	380 V / 50 Hz					
Temperature control accuracy	±1 °C					
Humidity control accuracy	±5%					
Filter class	G4					
Physical						
Weight (kg)	492	510	600	690	740	795
W × D × H (mm, excl. plenum)	1830 × 970 × 1950					
Up-flow plenum height (mm)	450					

Room-level AC · constant temperature & humidity – Outdoor unit

Indoor unit model	SK-FAJ050U(D)RH-DS	SK-FAJ060U(D)RH	SK-FAJ070U(D)RH	SK-FAJ080U(D)RH	SK-FAJ090U(D)RH	SK-FAJ100U(D)RH
Outdoor unit model	SK-CY0522		SK-CY0582		SK-CY0752	SK-CY0742
Quantity	2					
Electrical & piping (example: within 30 m)						
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz		
Max. horizontal distance	60 m (with extension kit)					
Max. height difference (outdoor unit above)	20 m					
Max. height difference (outdoor unit below)	5 m					
Physical						
Dimensions (mm)	1530 × 400 × 996		1785 × 400 × 955		2035 × 400 × 996	
Weight (kg)	128		140	158	160	185

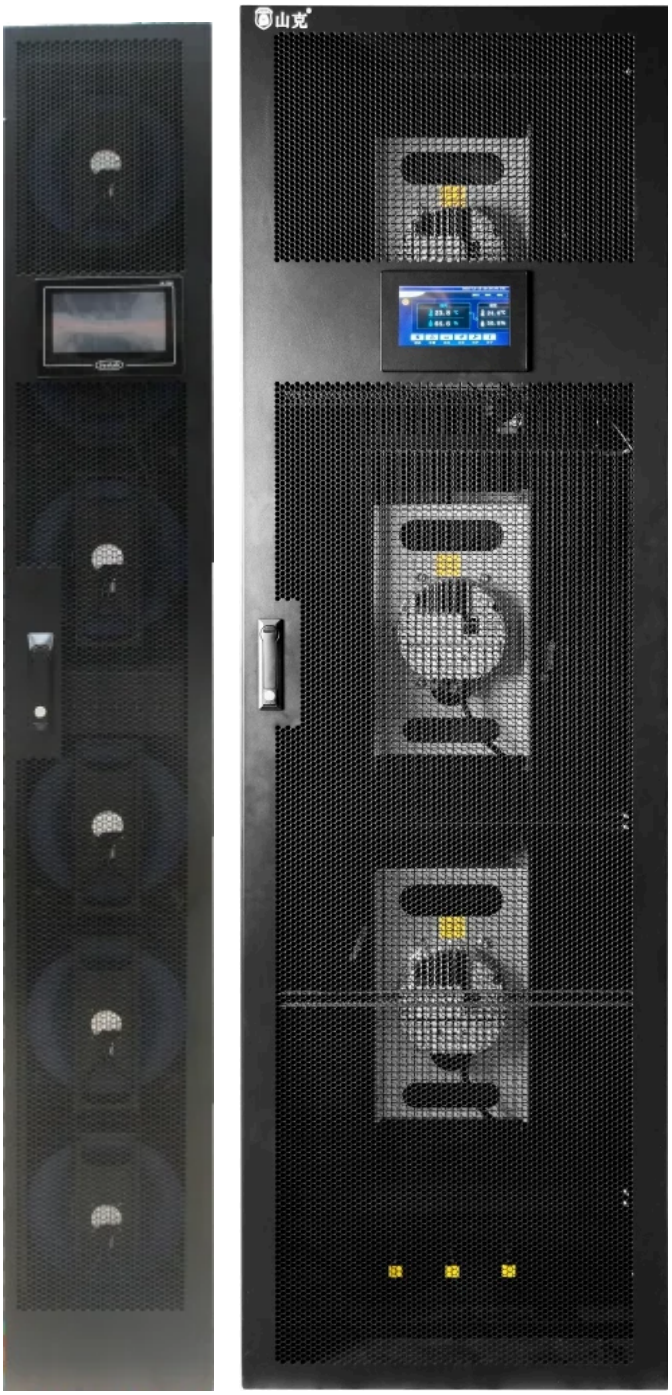
In-row precision AC (single circuit, 12.5–50 kW)

Overview

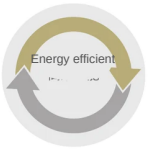
The LAJ in-row precision air conditioner is designed for high-heat-density rooms and micro-modules. It cools right next to the heat source, shortening supply and return air paths and raising the return air temperature; cooling output follows the load, with a high sensible heat ratio and high energy efficiency. A fully variable-speed design with a precise electronic expansion valve adapts to changing loads with 30–100% variable capacity and accurate temperature control. A 7-inch colour touch screen shows the status of every component, and up to 32 units can be group-controlled to keep the whole module stable. EC fan speed adjusts dynamically to the load, with several control modes for maximum energy savings. RS485 interface and leak alarm are standard.

Applications

· High-heat-density rooms
· Modular data centres
· Container data centres



Features



Energy efficient

· Sensible heat ratio of 1, cooling right at the heat source
· High-efficiency inverter compressor
· Eco-friendly R410A refrigerant
· Stepless-speed EC fans
· G4 air filters
· Variable-speed outdoor fan with stepless control



High reliability

· Designed for continuous 24/7, 365-day operation
· Automatic restart after power returns
· Complete automatic protection and alarms
· Fault self-diagnosis protects the unit on every front



Intelligent control

· Large 7-inch colour touch screen
· Temperature and humidity trend curves
· Standby, rotation, cascading and anti-competition operation
· RS485 interface and Modbus protocol as standard
· Networked group control of up to 32 units



Flexible configuration

· Several unit sizes
· Low-temperature start kit for the air-cooled condenser: operation down to −35 °C
· Customised cooling solutions available



Flexible configuration

Cold-aisle containment by design

A unique airflow design achieves cold-aisle containment without extra containment accessories.

Power protection

Protects the unit automatically on power faults and resumes normal operation when power returns.

Humidifier / heater

Handles room temperature and humidity without an additional room-level unit.

Rack-sized, IT-compatible

Compact and compatible with racks from different manufacturers, so it fits all kinds of data centres.

Leak detection

Optional leak detection avoids water hazards and keeps the system reliable.

Monitoring

Communication monitoring shows unit status, compressor and fan data and all alarm signals.



Technical specifications

In-row AC · cooling only – Indoor unit

Indoor unit model	SK-LAJ012F	SK-LAJ025F	SK-LAJ030F	SK-LAJ035F	SK-LAJ040F	SK-LAJ045F	SK-LAJ050F
System type	Single circuit, inverter						
Air supply	Horizontal supply						
Compressor	Inverter scroll						
Refrigerant	R410A						
Fan	EC centrifugal fan						
Expansion valve	Electronic expansion valve						
Heater type	—						
Humidifier type	—						
Performance							
Total cooling capacity (kW)	12.5	25.4	31	35.2	41	45.6	50.2
Sensible cooling capacity (kW)	12.5	25.4	31	35.2	41	45.6	50.2
Sensible heat ratio	1						
Airflow (m³/h)	3000	5200	6800	7800	9000	9500	10500
Heating capacity (kW)	—						
Humidification (kg/h)	—						
Condensate drain (mm)	20						
Pipe – gas line (mm)	16	22					
Pipe – liquid line (mm)	9.52	16					
Power supply (V/Hz)	380 V / 50 Hz						
Temperature control accuracy	±1 °C						
Humidity control accuracy	±5%						
Filter class	G4						
Physical							
Weight (kg)	158	185	263	270	272	280	
W × D × H (mm)	300 × 1100/1200 × 2000		600 × 1100/1200 × 2000				

In-row AC · cooling only – Outdoor unit

Indoor unit model	SK-LAJ012F	SK-LAJ025F	SK-LAJ030F	SK-LAJ035F	SK-LAJ040F	SK-LAJ045F	SK-LAJ050F
Outdoor unit model	SK-CY0191	SK-CY0522	SK-CY0582	SK-CY0662	SK-CY0742		SK-CY0893
Quantity	1						
Electrical & piping (example: within 30 m)							
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz			
Max. horizontal distance	60 m (with extension kit)						
Max. height difference (outdoor unit above)	20 m						
Max. height difference (outdoor unit below)	5 m						
Physical							
Dimensions (mm)	960 × 370 × 1260	1530 × 400 × 996	1785 × 400 × 955	1850 × 400 × 996	2035 × 400 × 996		2200 × 400 × 996
Weight (kg)	53	115	129	145	165	178	



Technical specifications

In-row AC · constant temperature & humidity – Indoor unit

Indoor unit model	SK-LAJ012FRH	SK-LAJ025FRH	SK-LAJ030FRH	SK-LAJ035FRH	SK-LAJ040FRH	SK-LAJ045FRH	SK-LAJ050FRH
System type	Single circuit, inverter						
Air supply	Horizontal supply						
Compressor	Inverter scroll						
Refrigerant	R410A						
Fan	EC centrifugal fan						
Expansion valve	Electronic expansion valve						
Heater type	PTC						
Humidifier type	Electrode						
Performance							
Total cooling capacity (kW)	12.5	25.4	31	35.2	41	45.6	50.2
Sensible cooling capacity (kW)	12.5	25.4	31	35.2	41	45.6	50.2
Sensible heat ratio	1						
Airflow (m³/h)	3000	5200	6800	7800	9000	9500	10500
Heating capacity (kW)	3		6				
Humidification (kg/h)	3						
Condensate drain (mm)	20						
Pipe – gas line (mm)	16	22					
Pipe – liquid line (mm)	9.52	16					
Power supply (V/Hz)	380 V / 50 Hz						
Temperature control accuracy	±1 °C						
Humidity control accuracy	±5%						
Filter class	G4						
Physical							
Weight (kg)	158	185	263	270	272	280	
W × D × H (mm)	300 × 1100/1200 × 2000		600 × 1100/1200 × 2000				

In-row AC · constant temperature & humidity – Outdoor unit

Indoor unit model	SK-LAJ012FRH	SK-LAJ025FRH	SK-LAJ030FRH	SK-LAJ035FRH	SK-LAJ040FRH	SK-LAJ045FRH	SK-LAJ050FRH
Outdoor unit model	SK-CY0191	SK-CY0522	SK-CY0582	SK-CY0662	SK-CY0742		SK-CY0893
Quantity	1						
Electrical & piping (example: within 30 m)							
Power supply (V/Hz)	220 V / 50 Hz			380 V / 50 Hz			
Max. horizontal distance	60 m (with extension kit)						
Max. height difference (outdoor unit above)	20 m						
Max. height difference (outdoor unit below)	5 m						
Physical							
Dimensions (mm)	960 × 370 × 1260	1530 × 400 × 996	1785 × 400 × 955	1850 × 400 × 996	2035 × 400 × 996		2200 × 400 × 996
Weight (kg)	53	115	129	145	165	178	

Rack-mount precision AC (single circuit, 3.5–12.5 kW)

Overview

The RMJ rack-mount air conditioner is a cabinet-level cooling unit designed for integrated cabinets, micro-modules and high-heat-density data centres. Installed in the rack right next to the heat source, it precisely removes the high sensible heat produced by the equipment, prevents local hot spots, shortens supply and return air paths and raises the return air temperature — improving efficiency, saving significant energy and supporting greener data centres.

Applications

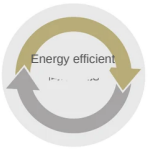
· Integrated cabinets
· Single-row cabinets
· Micro-modules
· Medium/high-heat-density rooms
· Modular data centres
· IT racks with blade servers





Rack-mount AC

Features



Premium components

· High-efficiency inverter compressor with ultra-precise PID control for tighter temperature control.
· Brand-name EC centrifugal fan adjusts airflow to demand in real time.
· Electronic expansion valve as standard, with smooth throttle adjustment for precise control.
· Chopper speed control of the outdoor motor over a wide 25–100% range.
· Eco-friendly R410A refrigerant.



Safe and reliable

· Very wide grid tolerance (±20% voltage) for harsh power conditions.
· Multiple drainage and anti-backflow design keeps water out of the rack.
· Intelligent refrigerant charge self-detection and early warning keep the system safe.



Highly intelligent

· High-efficiency inverter compressor with ultra-precise PID control.
· Proprietary vibration-compensation algorithm for wide-frequency operation.
· Reliable anti-condensation algorithm protects the servers.



Easy installation and maintenance

· Standard modular design fits standard server racks.
· Plug-in fans, plug-in control wiring terminals and a pull-out control box make installation and maintenance fast; minimal rack space used, 100% front maintenance.
· Refrigerant piping with quick couplings.

Options

· Several air supply options
· PTC electric heating
· Optional low-temperature kit for cold outdoor conditions
· Optional extension kit for long copper pipe runs
· Customised cooling solutions available



Technical specifications

Rack-mount AC · cooling only – Indoor unit

Indoor unit model	SK-RMJ003S	SK-RMJ007S	SK-RMJ012S
System type		Single circuit, inverter	
Air supply		Three-side supply	
Refrigerant		R410A	
Fan		EC centrifugal fan	
Expansion valve		Electronic expansion valve	
Heater type		—	
Humidifier type		—	
Performance			
Total cooling capacity (kW)	3.5	7.5	12.5
Sensible cooling capacity (kW)	3.5	7.5	12.5
Sensible heat ratio		1	
Airflow (m^3/h)	800	1650	2200
Heating capacity (kW)		—	
Humidification (kg/h)		—	
Condensate drain (mm)		20	
Pipe – gas line (mm)	9.52	16	
Pipe – liquid line (mm)	6.35	9.52	
Power supply (V/Hz)		220 V / 50 Hz	
Temperature control accuracy		±1 °C	
Humidity control accuracy		±5%	
Filter class		G2	
Physical			
Weight (kg)	25	35	43
W × D × H (mm)	443 × 715 × 218	443 × 715 × 354	443 × 715 × 442

Rack-mount AC · cooling only – Outdoor unit

Indoor unit model	SK-RMJ003S	SK-RMJ007S	SK-RMJ012S
Outdoor unit model	SK-CJT3005	SK-CJT3008	SK-CJT3013
Quantity		1	
Compressor		Inverter rotary	
Electrical & piping (example: within 30 m)			
Power supply (V/Hz)		220 V / 50 Hz	
Max. horizontal distance		30 m (with extension kit)	
Max. height difference (outdoor unit above)		5 m	
Max. height difference (outdoor unit below)		10 m	
Physical			
Dimensions (mm)	765 × 268 × 550	960 × 370 × 810	960 × 370 × 1260
Weight (kg)	31	59	80



Technical specifications

Rack-mount AC · constant temperature & humidity – Indoor unit

Indoor unit model	SK-RMJ003SRH	SK-RMJ007SRH	SK-RMJ012SRH
System type		Single circuit, inverter	
Air supply		Three-side supply	
Refrigerant		R410A	
Fan		EC centrifugal fan	
Expansion valve		Electronic expansion valve	
Heater type		PTC	
Humidifier type		Wetted media	
Performance			
Total cooling capacity (kW)	3.5	7.5	12.5
Sensible cooling capacity (kW)	3.5	7.5	12.5
Sensible heat ratio		1	
Airflow (m^3/h)	800	1650	2200
Heating capacity (kW)		0.8	
Humidification (kg/h)		1	
Condensate drain (mm)		20	
Pipe – gas line (mm)	9.52	16	
Pipe – liquid line (mm)	6.35	9.52	
Power supply (V/Hz)		220 V / 50 Hz	
Temperature control accuracy		±1 °C	
Humidity control accuracy		±5%	
Filter class		G2	
Physical			
Weight (kg)	25	35	43
W × D × H (mm)	443 × 715 × 218	443 × 715 × 354	443 × 715 × 442

Rack-mount AC · constant temperature & humidity – Outdoor unit

Indoor unit model	SK-RMJ003SRH	SK-RMJ007SRH	SK-RMJ012SRH
Outdoor unit model	SK-CJT3005	SK-CJT3008	SK-CJT3013
Quantity		1	
Compressor		Inverter rotary	
Electrical & piping (example: within 30 m)			
Power supply (V/Hz)		220 V / 50 Hz	
Max. horizontal distance		30 m (with extension kit)	
Max. height difference (outdoor unit above)		5 m	
Max. height difference (outdoor unit below)		10 m	
Physical			
Dimensions (mm)	765 × 268 × 550	960 × 370 × 810	960 × 370 × 1260
Weight (kg)	31	59	80